Incident report analysis

Summary	The organisation recently experienced a DoS attack. This was spotted when the organisations network services suddenly stopped responding due to a flood of ICMP packets. As a result, normal internal network traffic could not access any network resources. Upon investigation, it was found that a malicious actor had sent the flood of ICMP packets into the company's network through and unconfigured firewall. The incident took two hours to resolve. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.
Identify	The type of attack used by the malicious actor was an ICMP flood attack. An ICMP packet is used for diagnosing health and connectivity of the device and the connection between the sender and the device. An ICMP request is sent, which the server processes each time to send a response. Malicious actors can take advantage of this protocol to send a high number of requests to overload the network device or server and cause it to stop working. The ICMP Flood affected the companies internal network, exploiting the vulnerable, unconfigured firewall on the network. The attack affected the network bandwidth and CPU resources of the server.
Protect	The unconfigured firewall needs to be configured with new rules to limit the rate of incoming ICMP packets. ICMP packets are useful tools for diagnosis but they need to be limited to prevent ICMP flood attacks. The same can be done for SYN packets as they may also lead to similar flood attacks. Source IP address verification should also be configured on the firewall to check for spoofed IP addresses on incoming ICMP packets and harden the organisations network.

Detect	Network monitoring SIEM tools need to be in place to detect abnormal traffic patterns and IDS systems can be configured to automatically detect suspicious characteristics.
Respond	To respond to such attacks in the future, early detection through the use of packet sniffing tools and IDS\IPS systems is needed. This kind of attack can also quickly turn into a DDoS attack, using multiple botnets to execute the attack. Blocking single addresses manually may not be effective against this situation. An automated IPS system can be configured to block ICMP packets and addresses one a threshold is reached in a short amount of time.
Recover	The affected devices include the internal network of the organisation and network services. The goal of incident response is to protect private data and staff. Business recovery is to bring back critical systems. A denial of service attack may mean that no assets were stolen, the goal could have just been to disrupt business operations. As such, critical systems need to be recovered and reset with the configurations necessary for prevention and response. A playbook needs to be created for the response and recovery plans.

Reflections/Notes: